

Section 12.4: The Cross Product

Vectors and the Geometry of Space

MTH 310
Calculus III

The Cross Product — A New Kind of Product

The **dot product** takes two vectors and returns a **scalar** ($\vec{a} \cdot \vec{b} \in \mathbb{R}$).

Now we introduce a second way to multiply vectors:

The **cross product** $\vec{a} \times \vec{b}$ takes two vectors in $\mathbb{R}^3$ and returns a **vector** that is **perpendicular** to both $\vec{a}$ and $\vec{b}$.

- The cross product is **only defined in 3D** (not 2D!)
- It's used everywhere: torque, angular momentum, normal vectors to surfaces, area calculations

Quick Review: Determinants

Definition 1: 2×2 Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example 1

Compute the determinant $\begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix}$.

$$\begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix} = (2)(1) - (3)(4) = 2 - 12 = -10$$

Definition 2: 3×3 Determinant (Cofactor Expansion)

Expand along the **first row**:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Notice the **alternating signs**: $+$, $-$, $+$.

Example 2

Compute the determinant:

$$\begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & -3 \\ -2 & 5 & 2 \end{vmatrix}$$

Expand along the first row using the cofactor formula.

The Cross Product Formula

Definition 3: Cross Product

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, then:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Expanding along the first row:

$$\vec{a} \times \vec{b} = \vec{i}(a_2b_3 - a_3b_2) - \vec{j}(a_1b_3 - a_3b_1) + \vec{k}(a_1b_2 - a_2b_1)$$

The result is a **vector** (note the $\vec{i}, \vec{j}, \vec{k}$). This is a "symbolic" determinant — the first row contains vectors, not numbers.

Example 3: Computing a Cross Product

Example 3
Let $\vec{a} = \langle -1, 3, 4 \rangle$ and $\vec{b} = \langle 5, 2, 1 \rangle$. Find $\vec{a} \times \vec{b}$.

Set up the 3x3 determinant and expand along the first row.

Your Turn: Cross Product Practice

Example 4
Compute $\langle 1, 0, 2 \rangle \times \langle 3, 1, -1 \rangle$.

Set up the determinant and expand. Check your answer with a dot product.

Key Properties of the Cross Product

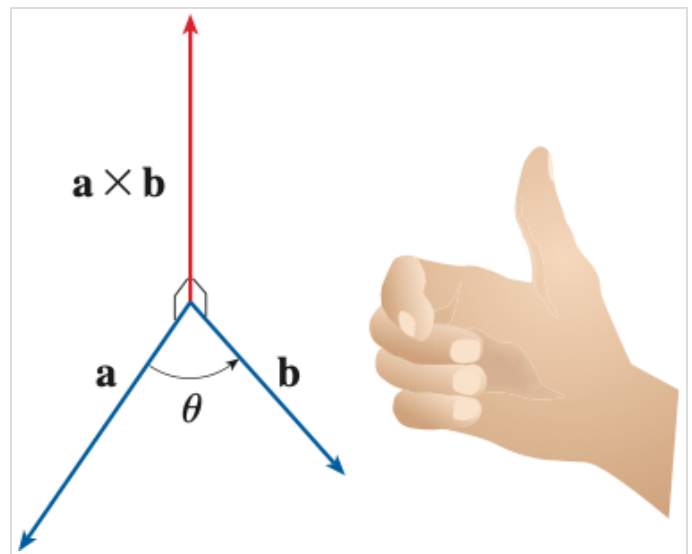
Here are the properties you'll use most often:

Property	Meaning
$\vec{a} \times \vec{a} = \vec{0}$	A vector is parallel to itself, so the cross product is the zero vector
$\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$	The result is perpendicular to both inputs
$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$	Anti-commutative — swapping order flips the sign

Order matters! $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ (they point in **opposite** directions).

Which direction does $\vec{a} \times \vec{b}$ point? Use the **right-hand rule**:

1. Point your fingers along $\vec{a}$
2. Curl them toward $\vec{b}$
3. Your **thumb** points in the direction of $\vec{a} \times \vec{b}$



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Magnitude of the Cross Product

Definition 4: Magnitude Formula

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \leq \theta \leq \pi$).

Compare with the dot product:

- **Dot product:** $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$ — measures **alignment**
- **Cross product magnitude:** $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$ — measures **perpendicularity**

If $\theta = 0$ or $\theta = \pi$ (vectors are **parallel**), then $\sin \theta = 0$, so:

$$\vec{a} \text{ and } \vec{b} \text{ are } \mathbf{parallel} \text{ if and only if } \vec{a} \times \vec{b} = \vec{0}.$$

This gives us two tests:

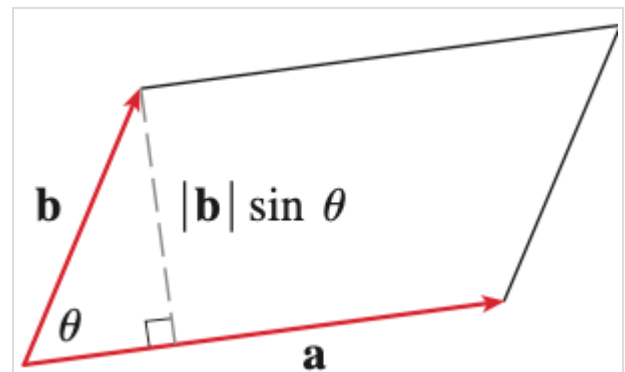
- **Parallel test:** $\vec{a} \times \vec{b} = \vec{0}$
- **Perpendicular test (from 12.3):** $\vec{a} \cdot \vec{b} = 0$

Area from the Cross Product

Recall that the area of a parallelogram is base $\times$ height:

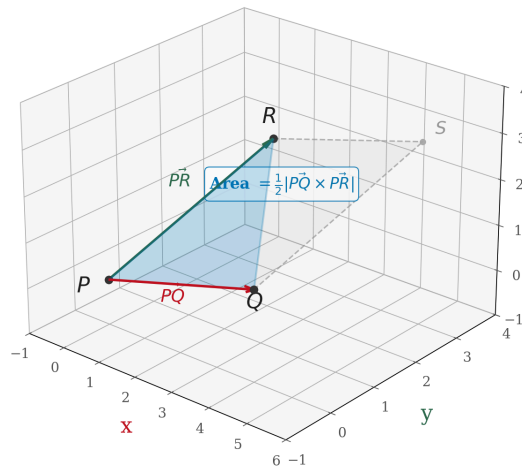
$$\text{Area} = |\vec{a}| (|\vec{b}| \sin \theta) = |\vec{a} \times \vec{b}|$$

- **Parallelogram** with sides $\vec{a}$ and $\vec{b}$: Area = $|\vec{a} \times \vec{b}|$
- **Triangle** with sides $\vec{a}$ and $\vec{b}$: Area = $\frac{1}{2} |\vec{a} \times \vec{b}|$



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This is incredibly useful: find the area of **any triangle in 3D** just from its three vertices!



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Example 5: Normal Vector and Area from Three Points

Example 5

Given $P(-2, 1, 7)$, $Q(-3, 4, 11)$, and $R(3, 3, 8)$:

- Find a vector perpendicular to the plane through P , Q , R .
- Find the area of triangle PQR .

Form two edge vectors from P , then take their cross product.

Cross Products of the Standard Basis Vectors

The standard basis vectors $\vec{i}$, $\vec{j}$, $\vec{k}$ follow a **cyclic pattern**:

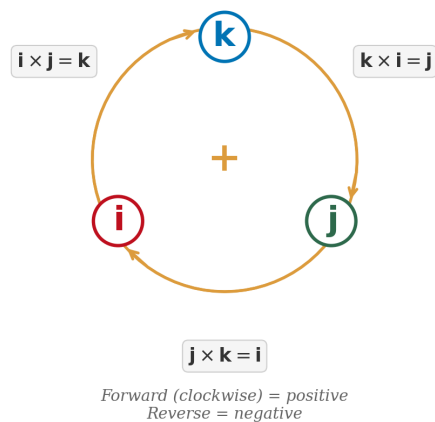
Forward (cyclic order $\vec{i} \rightarrow \vec{j} \rightarrow \vec{k} \rightarrow \vec{i}$):

$$\vec{i} \times \vec{j} = \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}$$

Backward (reversed order = negative):

$$\vec{j} \times \vec{i} = -\vec{k}, \quad \vec{k} \times \vec{j} = -\vec{i}, \quad \vec{i} \times \vec{k} = -\vec{j}$$

Remember the cyclic pattern: $\vec{i} \rightarrow \vec{j} \rightarrow \vec{k} \rightarrow \vec{i} \rightarrow \dots$
Going forward gives a positive result; going backward gives a negative.



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Homework

Section 12.4: 2, 3, 13, 16, 17, 19, 27, 29, 43, 44